

An Online Veterinary Appointment Management System

Requirement Engineering

1. User Panel (Pet / Patient)

User Requirement 1: The patient wants to create an account, log in, and manage their profile.

System Requirements:

1. The application should provide a 'Register' button for new bird owners/patients.
2. The application should provide a 'Log In' screen for existing patients.
3. A 'Manage Profile' section shall be available after logging in via the owner dashboard.

Functional Requirements:

1. The system shall allow new users to register an account.
2. The system shall provide secure authentication for patient login.
3. The system shall allow patients to manage their personal profile information.

User Requirement 2: The patient wants to find a doctor and book an appointment.

System Requirements:

1. The application shall display a list of available veterinarians (doctor list).
2. The application shall display individual doctor details and timing slots.
3. The booking interface shall provide options to choose a doctor, select a date, and select an available time slot.
4. If a specific time and date are already booked by another patient, the system shall mark the slot as not available.
5. Upon booking and entering payment details, a payment slip confirmation shall be displayed.

Functional Requirements:

1. The system shall list available doctors, their details, and their schedules.
2. The system shall allow patients to book an appointment by selecting a doctor, date, and valid time.
3. The system shall dynamically validate time slots to prevent double-booking.

4. The system shall generate a digital payment slip for the appointment transaction.

User Requirement 3: The patient wants to view and cancel their existing appointments.

System Requirements:

1. An 'Appointment History' screen should be available on the user panel.
2. Each upcoming appointment should have a 'view and cancel' button.

Functional Requirements:

1. The system shall allow patients to view their appointment history.
2. The system shall allow patients to view an appointment.
3. The system shall allow patients to cancel an appointment.

User Requirement 4: The patient wants to seek automated advice on bird health and disease prevention.

System Requirements:

1. An automated Chatbot interface shall be available on the patient panel.
2. The interface shall provide input options for users to ask questions regarding different bird diseases.

Functional Requirements:

1. The system shall process user queries through a chatbot interface.
2. The system shall answer questions regarding different types of diseases and how to prevent them.

User Requirement 5: The patient wants to cancel an existing appointment and receive a 70% payment refund.

System Requirements:

1. The system shall provide a 'Cancel' option for each booked appointment on the patient dashboard.
2. Upon confirming cancellation, the system interface shall prompt the user with details of the cancellation policy (30% fee deduction and 70% refund).
3. A viewable Refund Slip screen shall be displayed after successful cancellation.

Functional Requirements:

1. The system shall calculate and process a 70% refund of the paid amount upon successful appointment cancellation.
2. The system shall generate a refund acknowledgment slip/notice showing the deducted fee (30%) and the returned amount (70%).

User Requirement 6: The patient wants to view and print their digital medical prescription.

System Requirements:

1. A 'View Prescription' button shall be provided next to completed appointments in the patient panel.
2. A 'Print Prescription' option shall be available inside the prescription viewer interface.

Functional Requirements:

1. The system shall allow patients to view prescribed medications, diagnoses, and doctor instructions.
2. The system allows patients to download or print their digital prescriptions in PDF format.

2. Vet Panel (Doctor)

User Requirement 1: The doctor wants to log in and manage their availability.

System Requirements:

1. The Vet Panel shall have a secure login screen requiring assigned credentials.
2. A dashboard interface shall be provided to show patient information and schedules.
3. A 'Manage Availability' interface shall be provided.

Functional Requirements:

1. The system shall authenticate veterinarians using secure, assigned credentials.
2. The system shall track and display the doctor's physical routine schedule.
3. The system shall allow doctors to manage their availability schedule.

User Requirement 2: The doctor wants to view upcoming appointments, examine patients, issue prescriptions, and track daily time.

System Requirements:

1. A dashboard screen shall display a list of all patients booked for the doctor ("Show Patient").
2. The dashboard shall calculate and display statistics: how many patients have been seen, how many are remaining, and the remaining working time for the day.
3. A **'View and Treat'** button shall be available for each booked appointment.
4. A 'Patient Details & Prescription Generator' screen shall display deep insights into the selected patient's history and allow doctors to write and print prescriptions.

Functional Requirements:

1. The system shall display booked appointments, detailed patient profiles, and real-time daily queue metrics to the logged-in doctor.
2. The system shall allow doctors to treat patients, create digital prescriptions (including disease diagnosis, medicine details, and instructions), and save them to patient records.
3. The system shall allow doctors to print generated prescriptions directly from the treatment screen.

3. Admin Panel

User Requirement 1: The admin wants to manage doctors, patients, prescriptions, and appointment routines.

System Requirements:

1. A 'Vet Management' module shall be available with capability controls (vet manage).
2. A 'Patient Management' module shall be available with capability controls (patient manage).
3. A **'Prescription Management'** module shall be available with administrative control.
4. An appointment routine control center shall be available to set physical (phy) consultation boundaries.

Functional Requirements:

1. The system shall allow the admin to manage (add, update, or remove) veterinarians.
2. The system shall allow the admin to manage patient and owner data records.
3. The system shall allow the admin to manage, view, and audit all medical prescriptions generated within the platform.

4. The system shall provide infrastructure for assigning doctor appointment routines.
5. The system shall enforce role-based access control for all users.

User Requirement 2: The admin wants to monitor all appointments and generate reports.

System Requirements:

1. A dashboard shall be available to monitor all appointments.
2. A 'Reports' section should be available.
3. The 'Reports' section shall provide options to generate daily, weekly and monthly reports.

Functional Requirements:

1. The system shall allow the admin to monitor and control all appointments.
2. The system shall allow the admin to generate daily and weekly reports.
3. The system shall track appointment statuses (Pending, Confirmed, Completed) for monitoring.
4. The system shall track refund transactions and reflect the 70% returned amount in financial and weekly/monthly reports.

4. Non-Functional Requirements

Requirements:

1. **Portability:** The system should be a web-based application accessible seamlessly from both desktop and mobile web browsers.
2. **Security:** The system should provide secure authentication mechanisms across all user roles (Admin, Vet, Pet).
3. **Access Control:** The system shall enforce role-based access control to keep medical histories and administrative settings locked to unprivileged accounts.

4. **Database:** The system shall use relational or organized database tables for storing vet profiles, patient records, payment logs, chatbot intents, and scheduled appointments.
5. **Knowledge Base / NLP:** The chatbot should use a simple LLM API mapped to an avian disease dataset for accurate prevention answers.